A PROJECT REPORT ON
"TIC TAC TOE GAME IN PYTHON"

Submitted by:
Nitish Kumar
Roll No: 25MCA20292
Course: MCA (1st Semester)

Under the Guidance of:
Kawaljit kaur

Department of Computer Applications
Chandigarh University

BONAFIDE CERTIFICATE

Certified that this project report entitled “TIC TAC TOE GAME IN PYTHON” is the bonafide work of Mr. Nitish Kumar (Roll No: 25MCA20292), who carried out the project work under my supervision.

(Signature of Guide)

Ms. Kawaljit kaur
Department of Computer Applications
Chandigarh University

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ABSTRACT

This project titled 'Tic Tac Toe Game in Python' is a console-based two-player game that allows players to play the classic Tic Tac Toe using numbers 0–8 to mark positions. The project demonstrates the use of lists, functions, loops, and conditional statements in Python. It provides a simple and interactive environment for players.

AIM

To design and develop a simple two-player Tic Tac Toe game using Python programming language.

INTRODUCTION

Tic Tac Toe is one of the most popular games, often used to learn logic-based programming. This project is built using Python, where two players take turns marking spaces in a 3×3 grid. The player who places three marks in a horizontal, vertical, or diagonal row wins the game. The project runs on the command line and uses lists to store player moves.

BRIEF DESCRIPTION

This project uses Python lists to track the state of the board and logical checks to detect wins or draws. The printBoard() function displays the board, checkWin()

checks the winning condition, and the main loop alternates turns. The program ensures valid input and displays the final result.

OBJECTIVES

- To implement logical and condition-based programming.
- To understand the use of lists and functions in Python. - To create an interactive console-based game. - To apply loops and decision-making concepts practically.

ALGORITHM / FLOWCHART Algorithm:

1. Initialize two lists (xState, zState) with zeros.
2. Display the board using printBoard().
3. Ask the player to choose a position (0–8).
4. Validate input and mark the position.
5. Check for a winning condition using checkWin().
6. If no winner and all positions are filled, declare a draw.
7. Display final board and result.

PROGRAM CODE

```
def sum(a, b, c):
return a + b + c def
```

```
printBoard(xState,
zState):
   zero = 'X' if xState[0] else ('O' if zState[0] else 0)
one = 'X' if xState[1] else ('O' if zState[1] else 1)
two = 'X' if xState[2] else ('O' if zState[2] else 2)
three = 'X' if xState[3] else ('O' if zState[3] else 3)
four = 'X' if xState[4] else ('O' if zState[4] else 4)
five = 'X' if xState[5] else ('O' if zState[5] else 5)     six
= 'X' if xState[6] else ('O' if zState[6] else 6)     seven
= 'X' if xState[7] else ('O' if zState[7] else 7)     eight =
'X' if xState[8] else ('O' if zState[8] else 8)

   print(f"\n{zero} | {one} |
{two}")     print("--+---+--")
print(f"{three} | {four} | {five}")
print("--+---+--")
   print(f"{six} | {seven} | {eight}\n")
```

PROGRAM EXPLANATION

The project is built using Python functions and lists. The printBoard() function displays the grid. The checkWin() function verifies all possible winning positions. The xState and zState lists store each player's moves. The program alternates between X and O turns until a winner is found or a draw occurs.

SAMPLE OUTPUT

```
  Welcome to Tic Tac Toe!

0 | 1 | 2
--+---+--
3 | 4 | 5
--+---+--
6 | 7 | 8

X's Turn
Enter a position (0-8): 0

X | 1 | 2
--+---+--
3 | 4 | 5
--+---+--
6 | 7 | 8

O's Turn
Enter a position (0-8): 4

X | 1 | 2
--+---+--
3 | O | 5
--+---+--
6 | 7 | 8
```

X won the match!

CONCLUSION

The Tic Tac Toe Game in Python successfully demonstrates the use of basic programming concepts such as functions, loops, lists, and conditional statements. It is an engaging way to understand how to implement logic-based games using Python.

FUTURE SCOPE

- Add a computer (AI) opponent.
- Create a graphical user interface using Tkinter.
- Keep track of player scores.
- Allow replaying multiple rounds.

REFERENCES

1. www.geeksforgeeks.org
2. www.programiz.com
3. Python Official Documentation (docs.python.org)

GITHUB LINK

https://github.com/nitishprasher-debug/tic-tack-toe.git